# Symposium Program

MSS 2026
Mutational Scanning Symposium

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| | **Wednesday** | | | |
| | **March 25th** | | | |
| | *Time* | *Speaker* | *Institution* | *Title* |
| | **10:00-10:15** | **Welcome to Country** | | |
| **Session 1** | **10:15-11:00** | **Fumitaka Inoue** | **Kyoto University** | ***Deciphering human functional genome using Massively Parallel Reporter Assays*** |
| Chair: Irene Gallego Romero | 11:00-11:15 | Chanodya Pasansi Ranwala | University of Adelaide | Unraveling Regulatory Complexity: Functional Characterisation of Enhancer Variants associated with Common Epilepsy |
| | 11:15-11:30 | Asfar Lathif Salaudeen | The University of British Columbia | Expanding the gene editing toolkit to decipher endogenous causal variants |
| | **11:30-11:45** | **Coffee break** | | |
| **Session 2** | 11:45-12:00 | Daniel V Brown | Walter and Eliza Hall institute of Medical Research | Active learning of the sequence-function landscape in a far-red fluorescent protein |
| Chair: Lachlan Jolly | 12:00-12:15 | Maddy Comerford | University of Melbourne | Mapping the gene regulatory landscape of archaic hominin introgression in modern Papuans |
| | 12:15:12-30 | Justine Shih | Broad Institute of MIT and Harvard | Mapping Genetic Variation to Cellular Phenotypes with Targeted Optical Endogenous Sequencing |
| Lara Muffley | 12:30-12:45 | Bernd Willems | Twist Bioscience | Writing the Future of Mutational Scanning |
| | **12:45-13:45** | **Lunch + posters on your own** | | |
| **Session 3** | **13:45-14:15** | **Bryony Thompson** | **Royal Melbourne Hospital** | **Lessons Learned from Applying Functional Data in Real-World Clinical Variant Classification** |
| Chair: Mandy Spurdle | 14:15-14:30 | Melissa Gilbert | The Children's Hospital of Philadelphia & University of Pennsylvania | The clinical utility of Multiplexed Assays of Variant Effects (MAVEs) in a pediatric cohort |
| | 14:30-14:45 | Richard James | Seattle Children's Research Institute | Functional Assessment of Genetic Variants of CARD11 and BCL10 with Saturation Genome Editing |
| | 14:45-15:00 | Ben Capodanno | Brotman Baty Institute at the University of Washington | MaveMD: A functional data resource for genomic medicine |
| | 15:00-15:15 | Emmylou C. Nicolas-Martinez | Adelaide University | RNA variants resolved in non-expressed genes by transactivation |
| | **15:15-15:45** | **Coffee Break** | | |
| **Session 4** | **15:45-16:15** | **Alex Wagner** | **Nationwide Children's Hospital** | **From Functional Assays to Computable Knowledge: Making MAVE Data Usable in Variant Interpretation** |
| Chair: John Christodoulos | 16:15-16:30 | Steven E. Brenner | University of California, Berkeley | Findings from the Critical Assessment of Genome Interpretation (CAGI), seventh edition: a community experiment to evaluate phenotype prediction |
| | 16:30-16:45 | Ashley P.L. Marsh | Ambry Genetics | Clinical translation of a MUTYH MAVE into a diagnostic laboratory workflow enables large-scale VUS resolution |
| | 16:45-17:00 | Frederick P. Roth | University of Pittsburgh | Landscapes of human AAT missense-variant effects reveal pathogenic variation and genetic suppressors |

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| | **Thursday** | | | |
| | **March 26th** | | | |
| | *Time* | *Speaker* | *Institution* | *Title* |
| **Session 5:** | **9:00-9:45** | **Matthew Hurles** | **Wellcome Sanger Institute** | ***Keynote Speaker - Scaling MAVEs: are we at the end of the beginning?*** |
| **Chair: Rehan Villani** | **9:45-10:15** | **Chai-Ann Ng** | **Victor Chang Cardiac Research Institute** | **Integrating MAVE and patch clamp data to improve classification and risk prediction in KCNH2-LQTS** |
| | 10:15-10:30 | Andrew Glazer | Vanderbilt University Medical Center | Three Multimodal Assays of SCN5A Variant Function Inform Arrhythmia Risk Prediction |
| | 10:30-10:45 | Sujatha Jagannathan | University of Colorado Anschutz Medical Campus | Single-codon resolution mapping of nonsense-mediated mRNA decay via genomic stop codon scanning of LMNA |
| | 10:45-11:00 | Lea Starita | University of Washington | 260,000 variant effect measurements and novel calibration methods drive resolution of VUS |
| | **11:00-11:30** | **Coffee Break** | | |
| **Session 6:** | **11:30-12:00** | **Nilah Ioannidis** | **University of California, Santa Cruz** | **Modeling the impact of personal genome variation on molecular phenotypes** |
| Chair: Sandra Cooper | 12:00-12:15 | Omar Tariq | University of British Columbia | tfGPRA: A High Throughput Platform for Transcription Factor Characterization |
| | 12:15-12:30 | Vanessa Burns | Wellcome Sanger Institute | Saturation Genome Editing to Clarify Variant Effect within 5'UTRs of Neurodevelopmental Disorder Genes |
| | 12:30-12:45 | Lorenzo Vaccaro | Telethon Institute of Genetics and Medicine | Genotype-phenotype single-cell transcriptomics for massive parallel assessment of genetic variants |
| Lara Muffley | 12:45-13:00 | Daniel Zimmerman | Ambry Genetics | MAVE Progress Report, How Ambry Genetics Validates and Deploys High-Throughput Functional Assays |
| | **13:00-14:00** | **Lunch + posters on your own** | | |
| **Session 7:** | **14:00-14:30** | **Jian-Rong Yang** | **Sun Yat-sen University, Guangzhou** | **Phenotypic Mutations in Collision: Negative Epistasis Between Transcription Errors and Translation Errors Revealed by DMS** |
| Chair: Matthew Wakefield | 14:30-14:45 | Srivatsan Raman | University of Wisconsin-Madison | From Variant Maps to Functional Design: Using Deep Mutational Scanning and Machine Learning to Engineer Programmable Bacteriophages |
| | 14:45-15:00 | Guillaume Diss | Friedrich Miescher Institute | The genetic architecture of the human bZIP interaction network |
| | 15:00-15:15 | Jacob Purcell | Monash University | High-resolution functional assessment of SMAD4 variants |
| | 15:15-15:30 | Rosa De Santis | Telethon Institute of Genetics and Medicine | From bulk to single cell: benchmarking analytical tools for Deep Mutational Scanning experiments |
| | 15:30-16:00 | **Coffee Break** | | |
| **Session 8:** | **16:00-16:30** | **Amelie Stein (v)** | **University of Copenhagen** | **Variant consequences in context** |
| Chair: Sefi Rosenbluh | 16:30-16:45 | Xinyu Wu | Walter and Eliza Hall institute of Medical Research | Elucidate the active conformation of cytokine receptors using deep mutational scanning |
| | 16:45-17:00 | Polina V Polunina | University of Freiburg | Modeling Viral Evolution as Sequence Transitions: A Transformer Approach Using Time-Resolved SARS-CoV-2 Spike Data |
| Fritz Roth | 17:00-17:30 | Selected poster short talks | | |
| | **17:30-19:30** | **Catered poster session** | | |
| | **19:45-late** | **Social function at the Gertrude Hotel (148 Gertrude St, five minute walk)** | | |

| | Friday | | | |
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| | **March 27th** | | | |
| | *Time* | *Speaker* | *Institution* | *Title* |
| **Session 9:** | **9:00-9:45** | **Maitreya Dunham** | **University of Washington** | ***Keynote Speaker: From ARS1 to ZWF1: some adventures in deep mutational scanning*** |
| Chair: Alan Rubin | **9:45-10:15** | **Mark Dawson** | **Peter MacCallum Cancer Centre** | **Characterising the functional landscape of the human PRC2 complex by base editing at single cell resolution** |
| | 10:15-10:30 | Maximilian Stammnitz (v) | Centre for Genomic Regulation | The genetic architecture of allosteric plant hormone receptors |
| | 10:30-10:45 | Jun J Yang | St. Jude Children's Research Hospital | MAVE-informed Protein Language Models for Predicting Pharmacogenetic Variant Function at Scale |
| | 10:45-11:00 | Douglas M. Fowler | University of Washington | Biochemical profiling of ~315,000 MAP kinase pathway protein variants in human cells with LABEL-seq |
| | **11:00-11:30** | **Coffee Break** | | |
| **Session 10:** | **11:30-12:00** | **Chris Hahn** | **SA Pathology and University of South Australia** | **Deep mutational scanning to measure the transactivational effect of GATA2 and ERG variants to benefit clinical interpretation** |
| Chair: Ebony Matotek | 12:00-12:15 | Alistair Dunham | Wellcome Sanger Institute | Standardisation and joint analysis of 2,041 MAVEs |
| | 12:15-12:30 | Yu-Jen (Jennifer) Lin | University of California, Berkeley | Critical Assessment of Genome Interpretation (CAGI) 7 ARSA missense stability prediction challenge identifies computational advances over state-of-the-art variant impact predictors |
| | 12:30-12:45 | Linda Wijaya | The Kids Research Institute Australia & University of Western Australia | Unravelling the RASopathy Syndromes using iPSC-derived neural disease modelling |
| Lara | 12:45-13:00 | TBC | Illumina Inc | TBC |
| | **13:00-14:00** | **Lunch (no posters)** | | |
| **Session 11:** Chair: Convenors | **14:00-14:30** | **Melissa Call** | **Walter and Eliza Hall institute of Medical Research** | **Mutational profiling of SARS-CoV-2 PLpro exposes inhibitor escape routes** |
| Conveners | 14:30-14:45 | Awards and closing | | |
| | | Coffee Break | | |
| | 15:00-17:00 | **Workshop** | | Clinical Application Workshop |